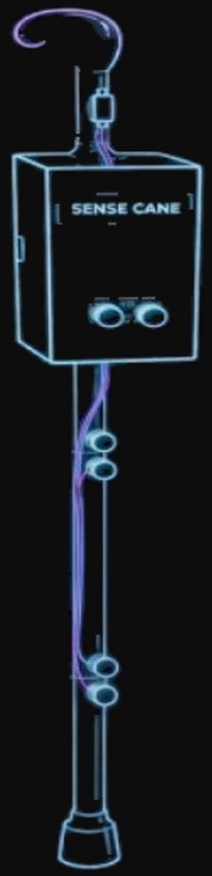



BUILD GUIDE · TECHNICAL OVERVIEW

# SENSE CANE

*ASSISTIVE TECHNOLOGY MADE ACCESSIBLE*

A PROJECT BY THE APRICITY FOUNDATION

# Components

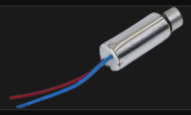
Everything you need to build the Sense Cane from scratch.

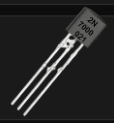
**01 Arduino Nano**
Main microcontroller

x1

**02 HC-SR04**
Ultrasonic distance sensor

x3

**03 Vibration Motor**
Coin-type, 3V-5V

x1

**04 N-ch MOSFET**
e.g. 2N7000 or similar

x1

**05 26650 Battery**
3.7V Li-ion

x1

**06 TP4056 Module**
Li-ion charging board

x1

**07 DC-DC Booster**
Step-up to 5V output

x1

**08 Schottky Diode**
e.g. 1N5819

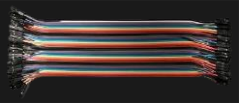
x1

**09 Rocker Switch**
Any SPST switch

x1

**10 400pt Breadboard**
MB-102 or equivalent

x1

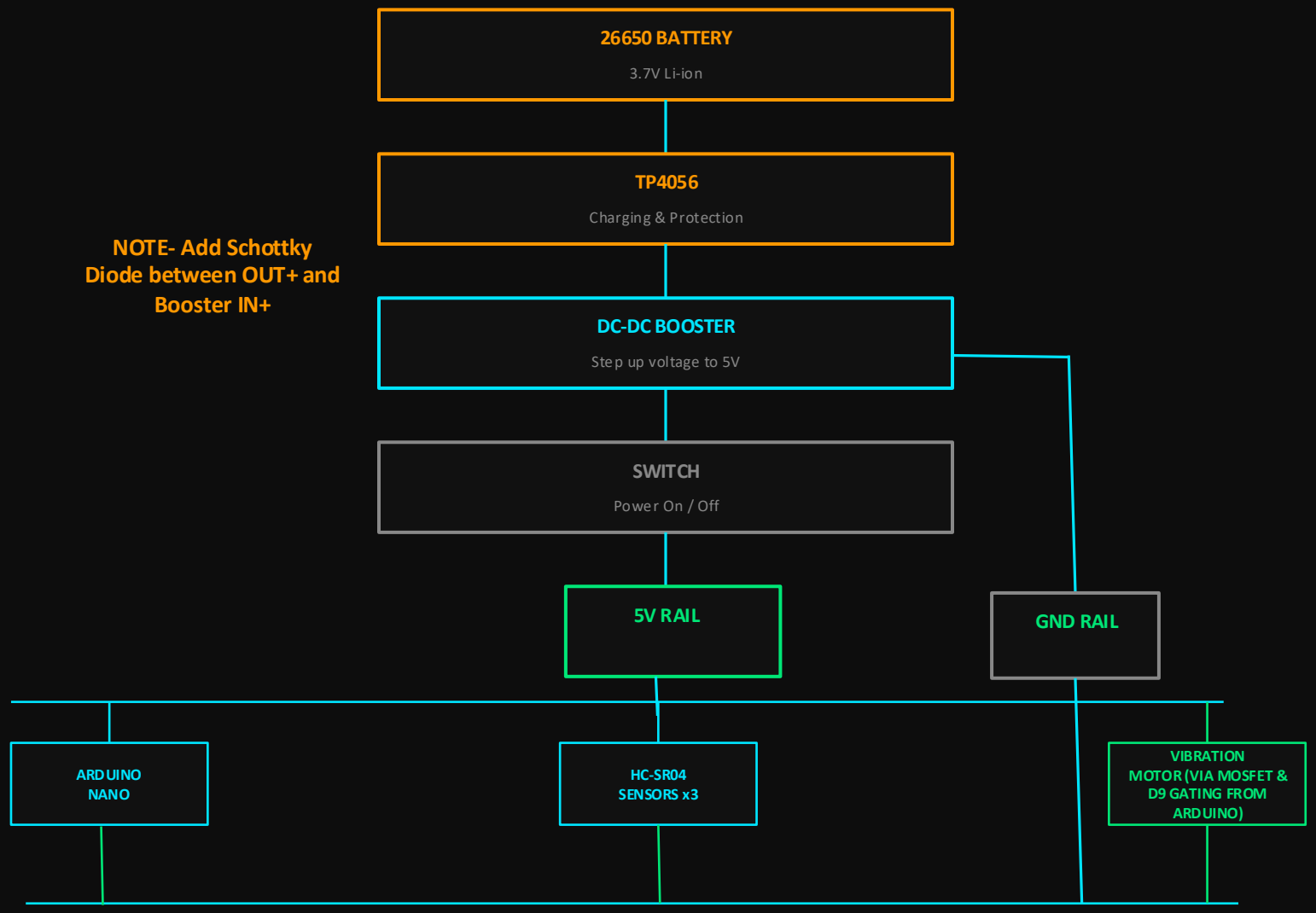
**11 Jumper Wires**
M-M, M-F assorted


**12 1k Ohm Resistor**
Optional — signal protection

x2

# Power Flow

An Overview of the overall power circuit:



# Pin Wiring

All connections between Arduino Nano and peripherals.

| Component | Pin | Arduino | Notes |
|---|---|---|---|
| HC-SR04 (High) | TRIG | D6 | |
| | ECHO | D7 | |
| HC-SR04 (Med) | TRIG | D4 | |
| | ECHO | D5 | |
| HC-SR04 (Low) | TRIG | D10 | |
| | ECHO | D11 | |
| Vibration Motor | Gate | D9 | Via MOSFET |
| All Sensors | VCC | 5V Rail | |
| All Sensors | GND | GND Rail | |
| Motor | VCC | 5V Rail | |
| Motor | GND | MOSFET D | Source→GND |

**MOTOR WIRING INSTRUCTIONS:**

+ from Motor → 5V rail
- from Motor → MOSFET Drain (D)
MOSFET Gate (G) → D9 (Arduino)
MOSFET Source (S) → GND rail
Diode → Make a Parallel Connection between + of Motor and – of Motor/Drain of MOSFET
Resistor → Connect Between D9 and Gate

# How the Code Works

A summary of SenseCane_Final.ino

**READ SENSORS**
Staggered 35ms to prevent crosstalk

**APPLY HYSTERESIS**
Buffer zone for obstacle detection (HC-SR04 gives constantly varying distance readings)

**EVALUATE COMBO**
Which sensor active?

**PLAY PATTERN**
Vibration Motor buzzes according to the Obstacle Scenario

*loop() repeats every 50ms*

### HYSTERESIS

Sensors arm below threshold.
Only disarm 15cm beyond it.
Kills edge flickering.

### STAGGERED READS

Sensors fire 35ms apart.
Prevents crosstalk between ultrasonic pulses.

### PATTERN-PACED

No timers. Pattern duration spaces repetitions naturally.
New obstacles trigger instantly.

## Tuning Reference

| | | |
|---|---|---|
| thresholdLow | 70cm | Increase if ground obstacles aren't caught early enough. |
| thresholdMed | 100cm | Decrease if too many false positives indoors. |
| thresholdHigh | 50cm | Keep tight — too large causes constant ceiling triggers. |
| hysteresis | 15cm | Increase if sensor still flickers at threshold edge. |
| buzzDuration | 200ms | Determines how long each Buzz lasts. |
| groupGap | 350ms | Gap between buzz groups in combo patterns. |